MSCS-634: Main Project: Final Report:

Sandesh Pokharel

University of the Cumberlands

MSCS-634 Advanced Big Data and Data Mining

Dr. Satish Penmatsa

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1. Introduction

The rapid growth of healthcare data has created new opportunities for applying data mining techniques to identify patterns and make informed decisions. In this project, we focused on heart disease prediction using a real-world medical dataset. Our goal was to use advanced data mining techniques—including classification, clustering, regression, and association rule mining—to uncover hidden patterns that could assist medical professionals in early diagnosis and decision-making.

This project was completed in multiple stages:

- **Deliverable 1:** Data understanding, preparation, and visualization.
- **Deliverable 2:** Implementation of regression and classification models.
- **Deliverable 3:** Application of clustering and association rule mining.

The final deliverable consolidates findings from all stages and presents insights, recommendations, and reflections on ethical and technical considerations.

2. Dataset Used and Rationale

We used the **Heart Disease dataset** derived from the Cleveland database, which is one of the most widely used and publicly available datasets in the UCI Machine Learning Repository. This dataset was chosen for several reasons:

- **Relevance:** Heart disease is one of the leading causes of death globally, making this dataset socially impactful.
- **Variety of Features:** It includes a mix of demographic (e.g., age, sex), clinical (e.g., cholesterol, resting blood pressure), and diagnostic (e.g., exercise-induced angina, ST depression) features.
- **Data Quality:** The dataset had a manageable size and relatively clean format, which was further refined during preprocessing.
- **Machine Learning Suitability:** The data includes both categorical and numerical features, making it ideal for experimenting with different machine learning models.

📌 **Screenshot showing the .head() output of the cleaned Heart Disease dataset (df.head())**

In [1]:
```python
# Import required libraries
import pandas as pd
import numpy as np

# Load the dataset
df = pd.read_csv("heart_disease_uci.csv")

# Display the first 5 rows
df.head()
```

Out[1]:

	id	age	sex	dataset	cp	trestbps	chol	fbs	restecg	thalch	exang	oldpeak	slope	ca	
0	1	63	Male	Cleveland	typical angina	145.0	233.0	True	lv hypertrophy	150.0	False	2.3	downsloping	0.0	
1	2	67	Male	Cleveland	asymptomatic	160.0	286.0	False	lv hypertrophy	108.0	True	1.5	flat	3.0	
2	3	67	Male	Cleveland	asymptomatic	120.0	229.0	False	lv hypertrophy	129.0	True	2.6	flat	2.0	reve
3	4	37	Male	Cleveland	non-anginal	130.0	250.0	False	normal	187.0	False	3.5	downsloping	0.0	
4	5	41	Female	Cleveland	atypical angina	130.0	204.0	False	lv hypertrophy	172.0	False	1.4	upsloping	0.0	

3. Key Insights from Data Preprocessing, EDA, and Feature Engineering

The preprocessing and exploratory data analysis (EDA) steps played a critical role in preparing the dataset for effective modeling. Below are the key activities and insights derived from these steps:

3.1 Data Preprocessing

We started by cleaning the dataset to ensure consistency and reliability. The following steps were taken:

- **Null Value Handling:**

Minimal missing data was present, and it was either dropped or imputed based on context.

- **Irrelevant Columns Removed:**

Features such as id were removed as they had no predictive value.

- **Categorical Encoding:**

Categorical variables like cp (chest pain type), thal, and restecg were encoded using LabelEncoder or OneHotEncoding.

- **Normalization/Scaling:**

Numerical columns like chol, thalach, and oldpeak were scaled using StandardScaler to ensure uniformity across features.

📌 **Screenshot :**

screenshot from Deliverable 1 showing the output of missing value checks and encoded data (df.isnull().sum(), df.dtypes, or encoded sample)

```
plt.show()

Missing values per column:

ca          611
thal        486
slope       309
fbs          90
oldpeak      62
trestbps     59
thalch       55
exang        55
chol         30
restecg       2
dtype: int64
```

Check for Remaining Missing Values

After performing data imputation, it's important to double-check if any missing values still exist in the dataset.

```
In [12]: # Show columns with remaining missing values and their counts
         df.isnull().sum()[df.isnull().sum() > 0]

Out[12]: sex      920
         fbs      920
         exang    920
         dtype: int64

In [13]: # Re-convert 'sex' column
         df['sex'] = df['sex'].replace({'Male': 1, 'Female': 0})

         # Re-convert 'fbs' column
         df['fbs'] = df['fbs'].astype(str).replace({'True': 1, 'False': 0})

         # Re-convert 'exang' column
         df['exang'] = df['exang'].astype(str).replace({'True': 1, 'False': 0})

         # Now check if anything is missing
         print(df.isnull().sum()[df.isnull().sum() > 0])

         sex    920
         dtype: int64

In [14]: # Fix 'sex' column — assume anything that's not 1 is 0 (since it should be binary)
         df['sex'] = df['sex'].apply(lambda x: 1 if str(x).lower() == 'male' or x == 1 else 0)

         # Now check again
         print(df.isnull().sum()[df.isnull().sum() > 0])

         Series([], dtype: int64)
```

3.2 Exploratory Data Analysis (EDA)

We visualized and analyzed feature distributions and relationships:

- **Age and Target Relationship:**

Found that the likelihood of heart disease increased with age, especially beyond 50.

- **Sex and Heart Disease:**

Males had a slightly higher occurrence rate of heart disease compared to females.

- **Cholesterol and Chest Pain Type:**

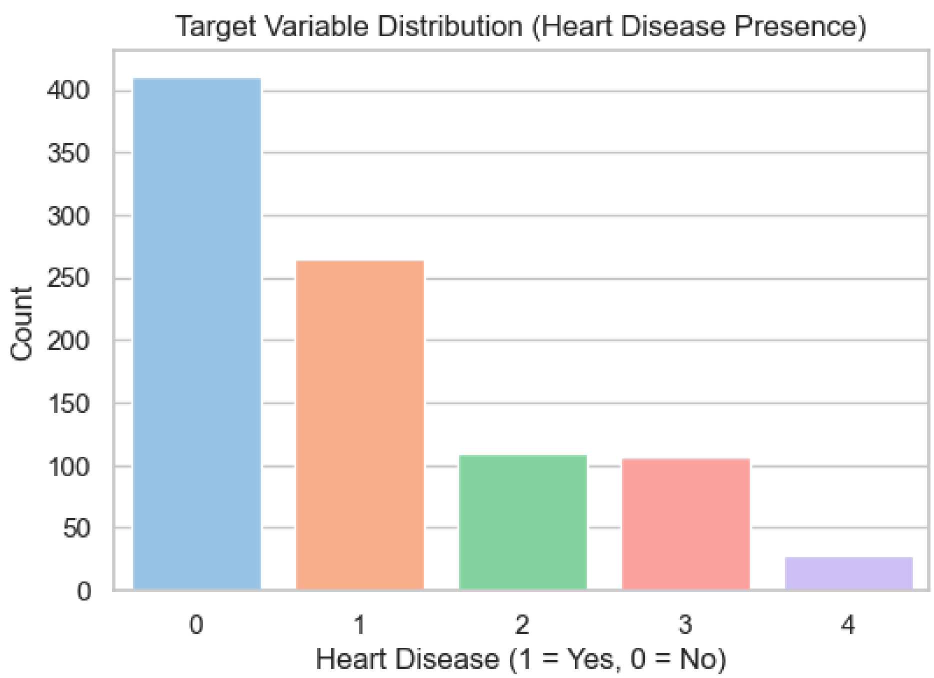
Patients with higher cholesterol levels tended to experience more severe chest pain types.

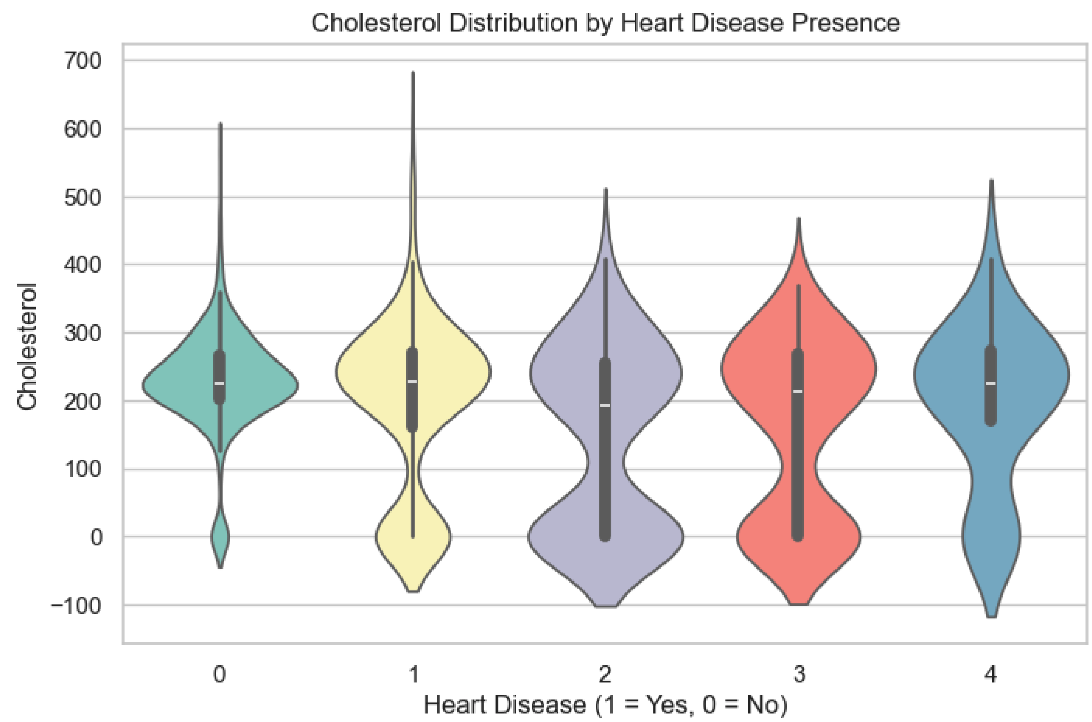
- **Thalach (Max Heart Rate):**

A lower maximum heart rate correlated with higher risk of heart disease.

📌 **Screenshot :**

bar chart or pair plot from Deliverable 1 showing relationships like Age vs. Target, Sex vs. Target, or Thalach distribution.





3.3 Feature Engineering

To enhance model performance:

- **Binning:**

Continuous features such as age and cholesterol were binned into categorical ranges to improve interpretability for some models.

- **Feature Selection:**

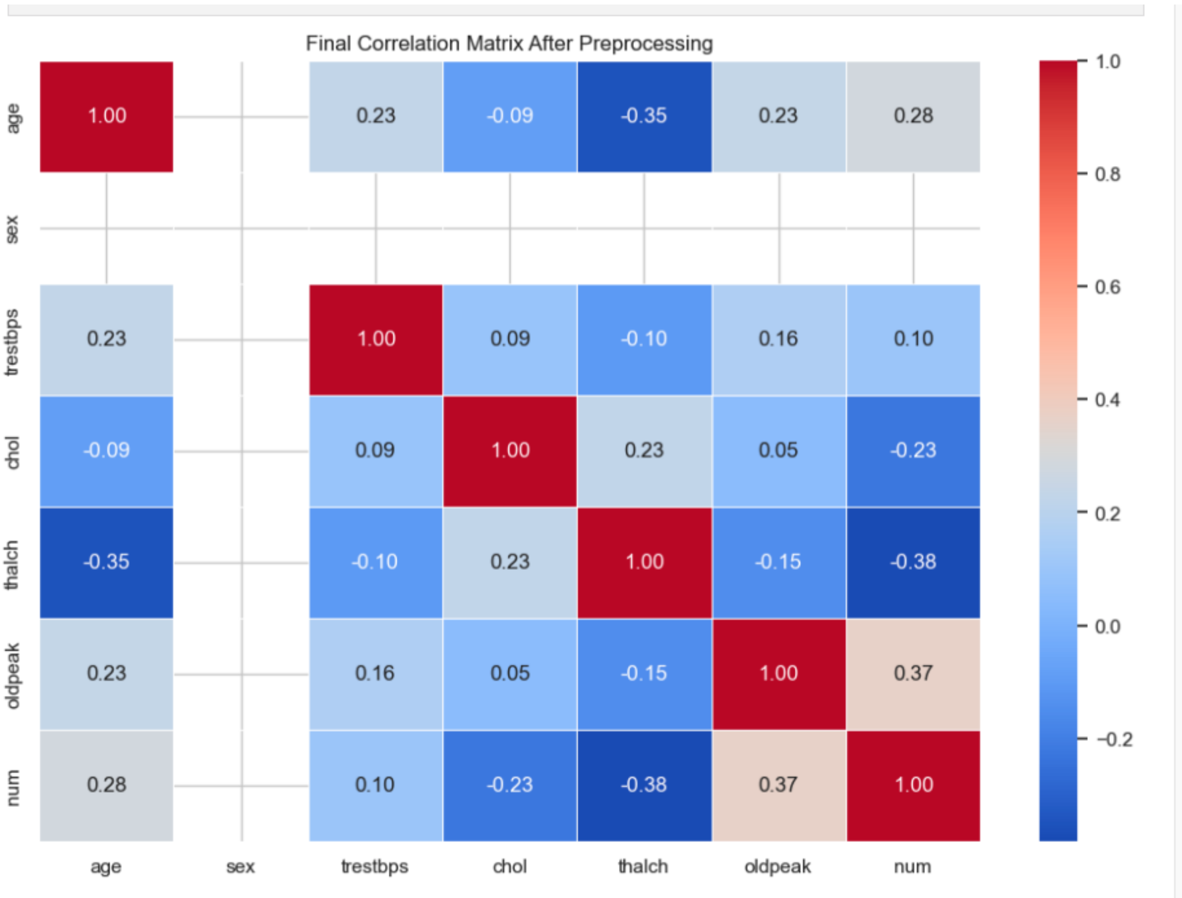
Features were selected based on correlation analysis and relevance to heart disease.

- **Derived Features:**

A few composite or transformed features (like age_group) were created to test whether grouped behavior added clarity.

📌 **Screenshot:**

correlation heatmap or feature importance plot from Deliverable ½



4. Results from Regression, Classification, Clustering, and Association Rule Mining

Our analysis incorporated a variety of supervised and unsupervised learning techniques to uncover patterns and make predictions based on the Heart Disease dataset.

4.1 Classification Models

We applied multiple classification algorithms to predict the presence of heart disease (target column). Models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

Models Used:

- **Logistic Regression:**

A baseline model that performed decently, especially for linearly separable data.

- **K-Nearest Neighbors (KNN):**

Tuned using GridSearchCV. Showed improved accuracy with appropriate k value.

- **Decision Tree Classifier:**

Delivered high interpretability and decent performance but risked overfitting.

- **Best Performing Model:**

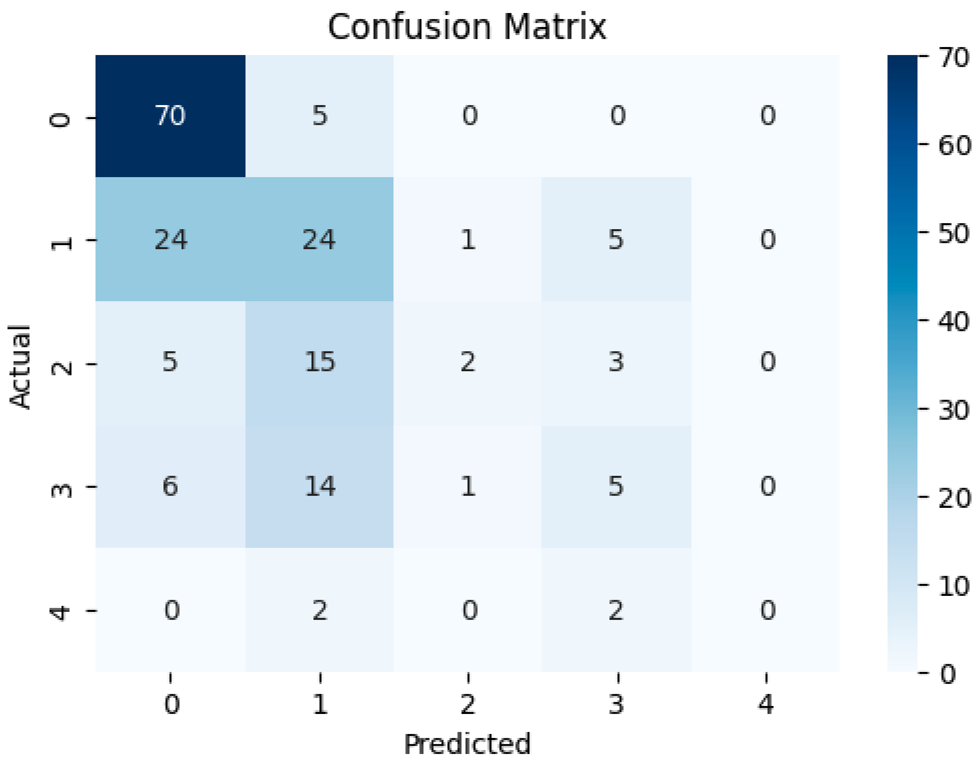
Based on overall metrics, KNN slightly outperformed others after tuning.

📌 **Screenshot :**

classification report comparison and confusion matrix plots from Deliverable 3.

Classification Report:

	precision	recall	f1-score	support
0	0.67	0.93	0.78	75
1	0.40	0.44	0.42	54
2	0.50	0.08	0.14	25
3	0.33	0.19	0.24	26
4	0.00	0.00	0.00	4
accuracy			0.55	184
macro avg	0.38	0.33	0.32	184
weighted avg	0.50	0.55	0.49	184



4.2

Clustering Analysis

We used clustering to explore natural groupings in the data without prior labels.

Methods Applied:

- **K-Means Clustering:**

Used the Elbow Method to find the optimal number of clusters. Patients were grouped based on similarities in features such as age, cholesterol, and thalach.

- **DBSCAN (Density-Based Spatial Clustering):**

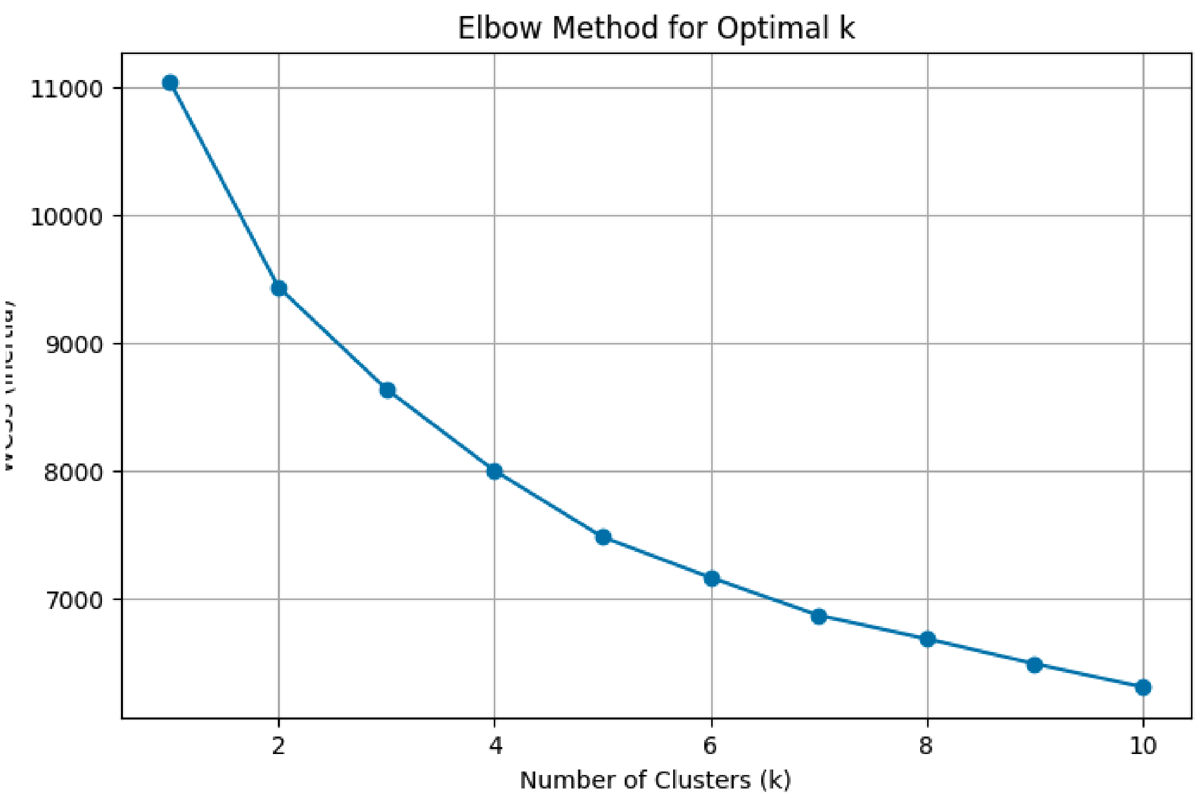
Effective in identifying outliers and forming arbitrary-shaped clusters. Best suited for datasets where noise and density variation are present.

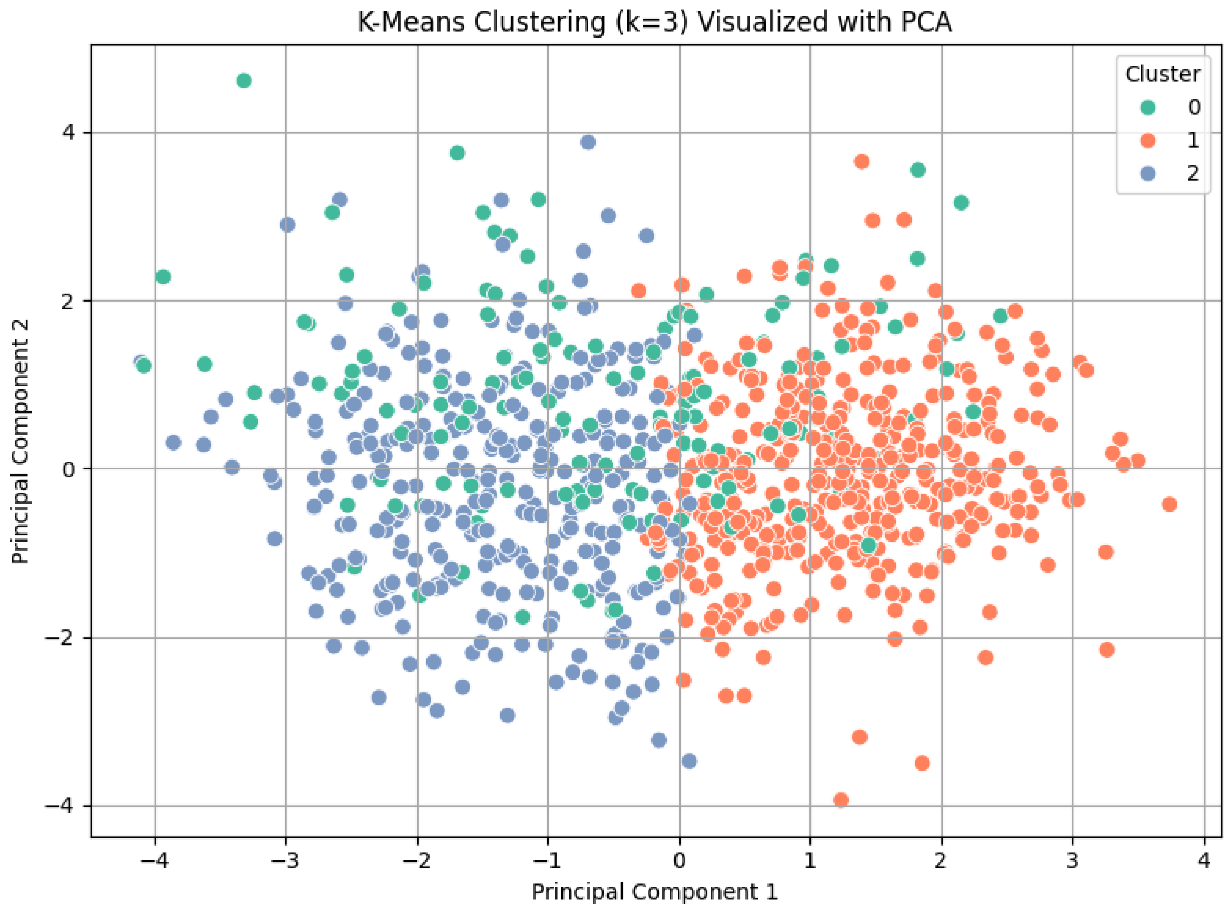
Key Insight:

Clusters revealed potential patient subgroups—e.g., low-risk younger patients vs. high-risk older patients with low heart rate.

📌 **Screenshot :**

K-Means elbow curve and 2D PCA scatter plot or





4.3 Association Rule Mining

We explored relationships between symptoms and diagnostic features using:

- **Apriori Algorithm**
- **FP-Growth Algorithm**

Notable Rules Found:

- Patients with typical angina and normal ECG had a high confidence of no heart disease.
- Patients with asymptomatic chest pain and fixed defect thal had a high lift for heart disease.

Support, Confidence, and Lift metrics were used to evaluate the strength of each rule.

📌 **Screenshot :**

sample association rules table and bar plot of top rules sorted by lift.

```
Top Association Rules:
```

	antecedents	consequents	support	confidence	lift
275	(thal_normal, num, restecg_normal, dataset_Hun...	(slope_flat)	0.080435	1.000000	1.406728
369	(dataset_Switzerland, slope_flat, cp_asymptoma...	(num)	0.066304	1.000000	1.807466
210	(num, restecg_normal, dataset_Hungary)	(slope_flat)	0.093478	1.000000	1.406728
493	(restecg_normal, cp_asymptomatic, dataset_Swit...	(num)	0.051087	1.000000	1.807466
388	(num, cp_asymptomatic, dataset_Hungary, thal_n...	(slope_flat)	0.065217	1.000000	1.406728
323	(num, restecg_normal, cp_asymptomatic, dataset...	(slope_flat)	0.075000	1.000000	1.406728
131	(num, dataset_Hungary)	(slope_flat)	0.114130	0.990566	1.393457
146	(restecg_normal, cp_asymptomatic, dataset_Hung...	(slope_flat)	0.105435	0.989796	1.392373
165	(thal_normal, num, dataset_Hungary)	(slope_flat)	0.100000	0.989247	1.391602
196	(thal_normal, restecg_normal, cp_asymptomatic,...	(slope_flat)	0.094565	0.988636	1.390742

5. Practical Recommendations Based on Findings

Based on the analysis of the Heart Disease dataset, several actionable insights and practical recommendations can be derived to support both clinical decision-making and data-driven healthcare initiatives.

5.1 Clinical Recommendations

- **Risk Stratification:**

Classification models, especially KNN and Decision Tree, were effective in predicting patients with a high risk of heart disease. These models can be integrated into early screening tools in primary care settings to flag high-risk patients based on clinical features such as age, cholesterol level, chest pain type, and maximum heart rate.

- **Targeted Interventions:**

Clustering analysis revealed distinct patient groups. For example, clusters with older patients and lower thalach levels had higher risk. Healthcare providers can tailor treatment plans and follow-up strategies based on these groupings.

- **Preventive Focus:**

Association rules highlighted strong patterns between asymptomatic chest pain and fixed defects. This suggests the need for routine checks even in asymptomatic individuals with other risk indicators.

5.2 Operational Recommendations

- **Model Integration in EHR Systems:**

The best-performing models can be integrated into hospital Electronic Health Record (EHR) systems to automate risk prediction during routine checkups.

- **Continuous Model Improvement:**

Periodic retraining with updated patient data is recommended to ensure model accuracy and relevance over time.

- **Outlier Detection for Monitoring:**

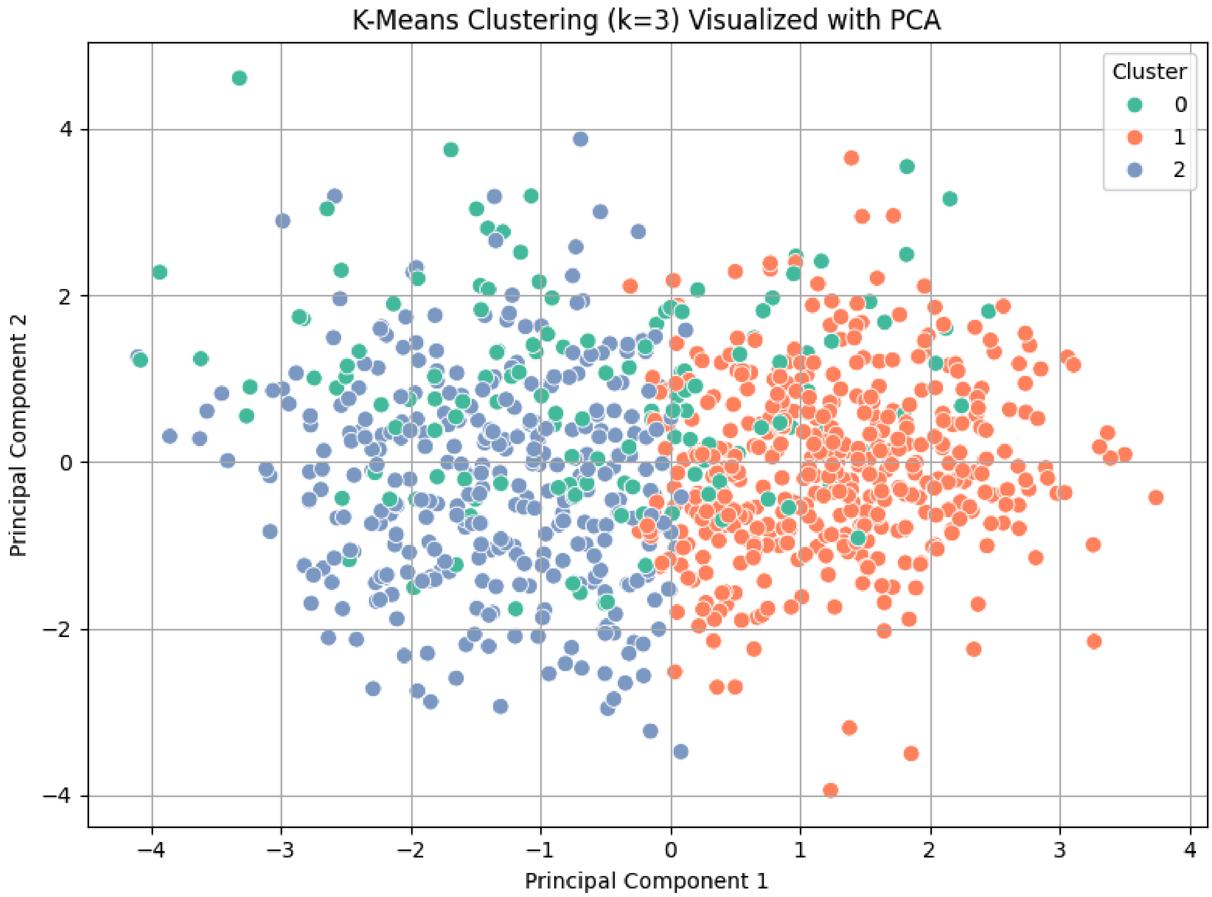
DBSCAN's ability to detect noise and outliers can be used for real-time alerts or monitoring unusual patient cases.

5.3 Visualization & Communication

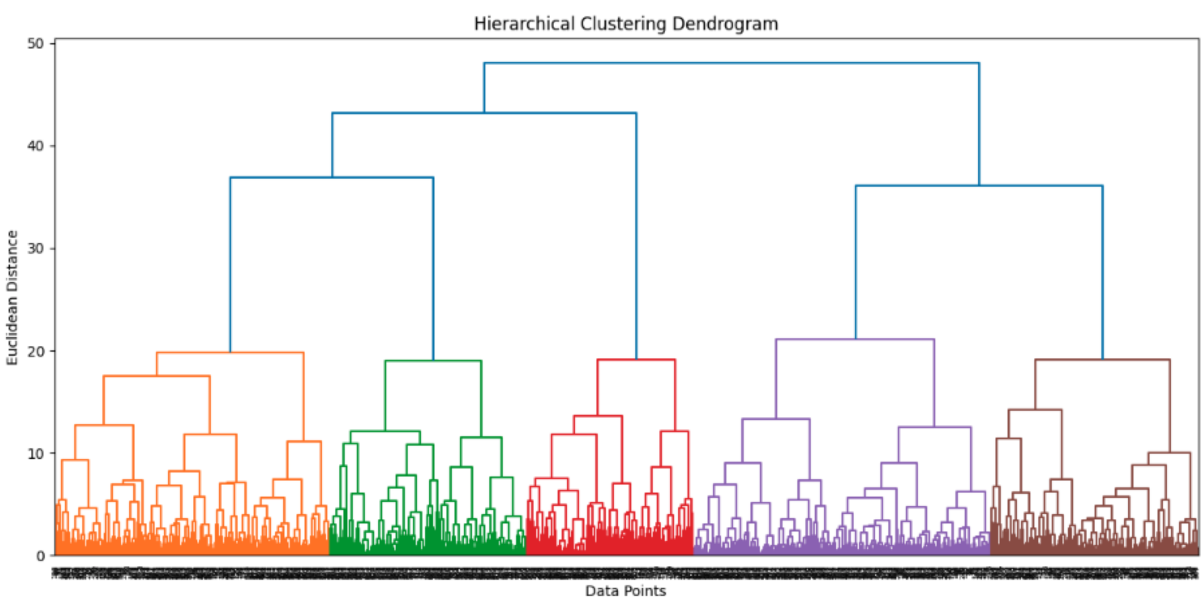
- Visualizations like heatmaps, scatter plots, and bar charts can be used in dashboards for clinicians to quickly interpret patient risk profiles.
- Association rule visualizations help communicate diagnostic rule logic to both clinical and non-technical stakeholders.

📌 **Screenshot :**

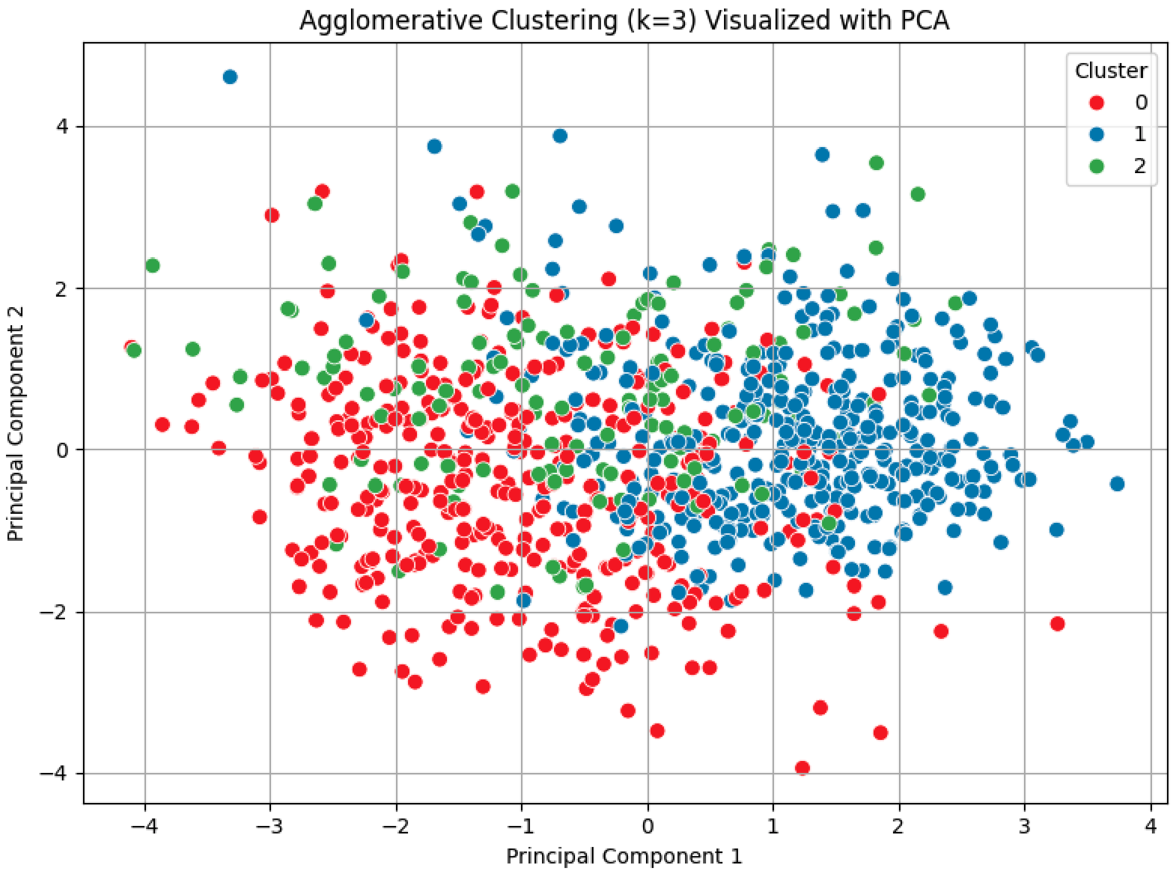
cluster visualization plot with annotated high-risk groups.



K-Means Clustering (k=3) Visualized with PCA
Cluster
0
1
2
Principal Component 1
Principal Component 2



Hierarchical Clustering Dendrogram
Euclidean Distance
Data Points



6. Ethical Considerations and Bias Mitigation

Ethical awareness is essential in any data science project, especially in healthcare, where outcomes can directly impact people's lives. Throughout this project, several ethical issues were considered and addressed.

6.1 Data Privacy and Anonymity

- The Heart Disease dataset used in this project is publicly available and anonymized, ensuring that no personally identifiable information (PII) was included.
- Nonetheless, we treated the data with the same caution as real-world patient data to maintain good practices.
- No external datasets were merged to prevent accidental re-identification risks.

6.2 Fairness and Bias

- **Feature Imbalance:**

The dataset showed potential imbalance in certain features like gender and age distribution. This can lead to models that unintentionally favor or penalize certain demographic groups.

- **Bias in Modeling:**

To reduce model bias:

  - We used **stratified sampling** when splitting training and test sets.
  - Performance metrics like **F1-score** were prioritized over accuracy to handle class imbalance.
  - Feature importance was reviewed to ensure that sensitive attributes (e.g., sex, age) were not overly dominating model predictions.

6.3 Interpretability and Transparency

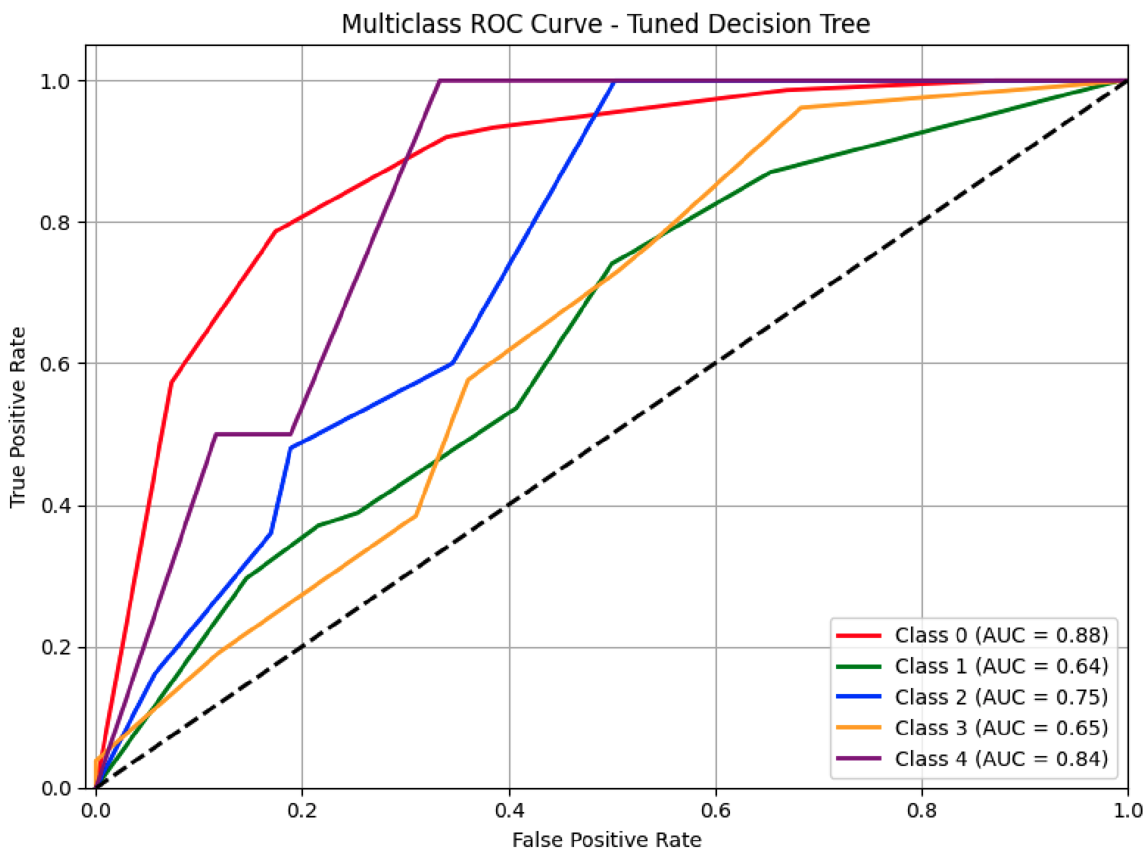
- **Model explainability** was prioritized using interpretable algorithms like Decision Trees and KNN instead of black-box models.
- Visualizations such as confusion matrices and ROC curves were used to communicate model behavior clearly to non-technical users.

6.4 Risk of Misuse

- It is important to note that while our models show good predictive power, they should not be used in isolation for medical diagnosis.
- These models are **decision-support tools** and should always be used in conjunction with expert clinical judgment.

📌 **Screenshot:**

- *confusion matrix and ROC curve visualizations for transparency illustration.*




```
Confusion Matrix:
[[69  5  0  1  0]
 [22 21  0 11  0]
 [ 3 17  0  5  0]
 [10 11  0  5  0]
 [ 2  0  0  2  0]]
```

7. Visualizations Supporting Key Conclusions

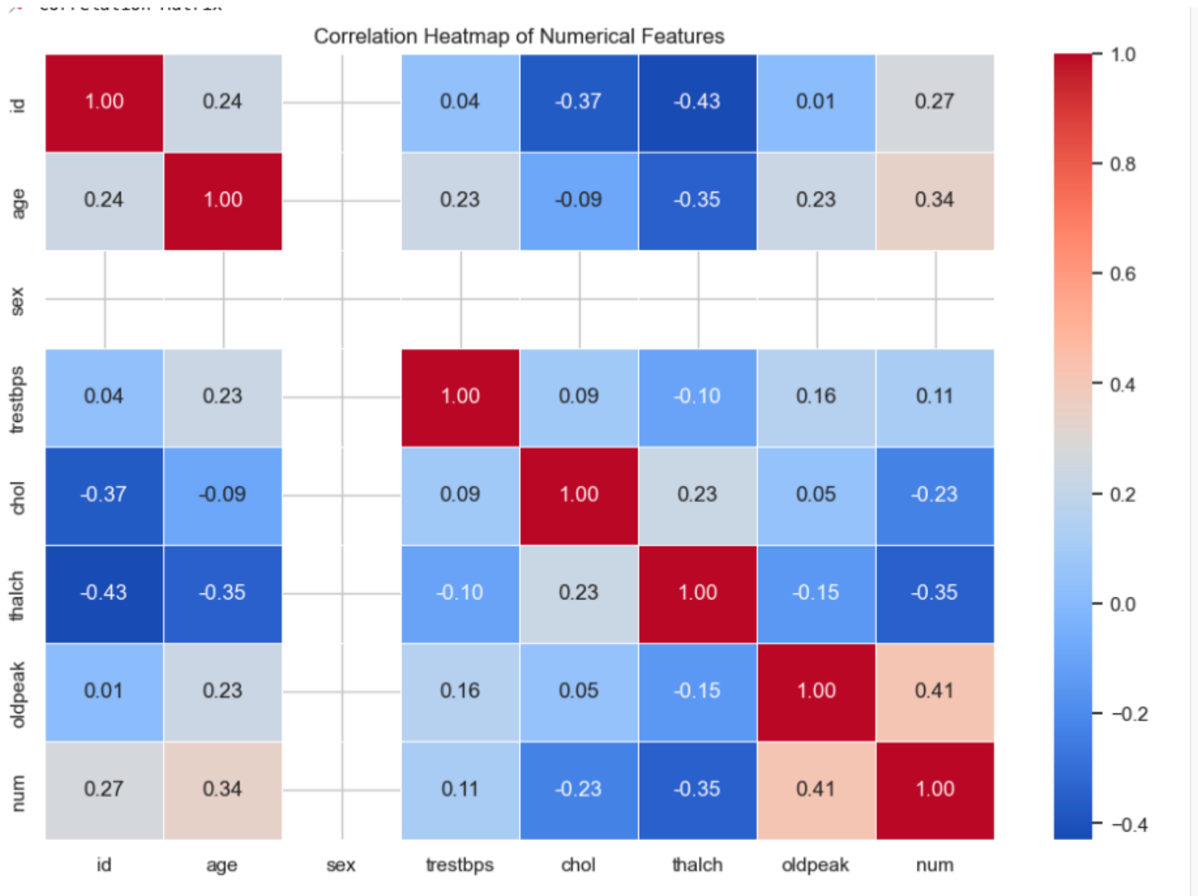
Visual storytelling played a crucial role throughout this project, helping interpret model performance and dataset characteristics more intuitively. Here are the key types of visualizations used and how they supported our findings:

7.1 Exploratory Data Analysis (EDA) Visuals

- **Correlation Heatmap:**

Helped us identify the strength of relationships between variables. Features such as cp (chest pain type), thalach (maximum heart rate), and exang (exercise-induced angina) were found to have strong correlation with the target variable.

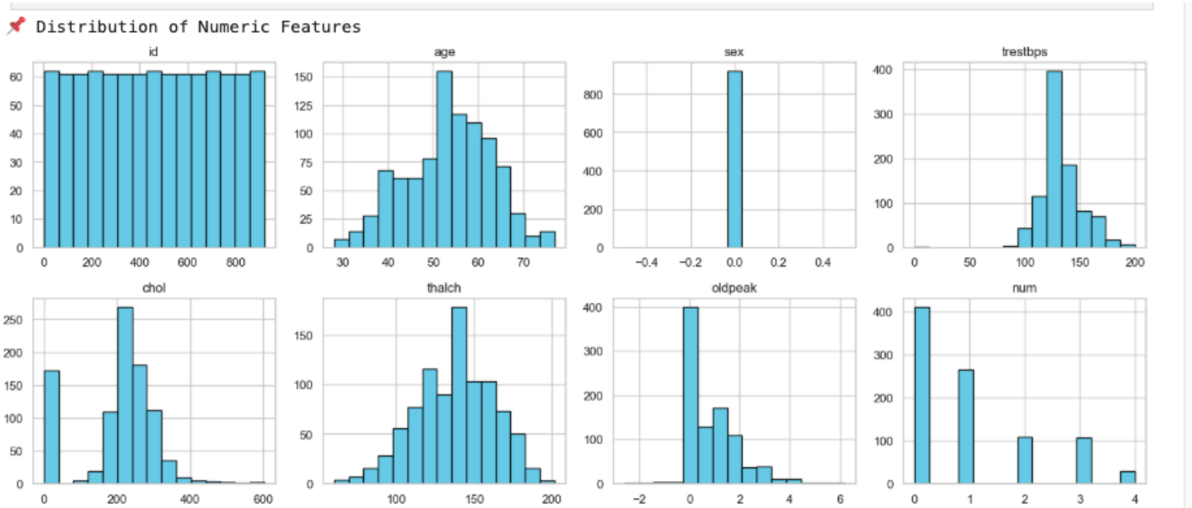
📌 *Screenshot of Correlation Heatmap*



- **Histogram and Boxplots:**

Provided insights into the distribution of numerical features and revealed outliers in columns such as chol (cholesterol) and thalach.

📌 *Screenshot of Histograms for Key Features*

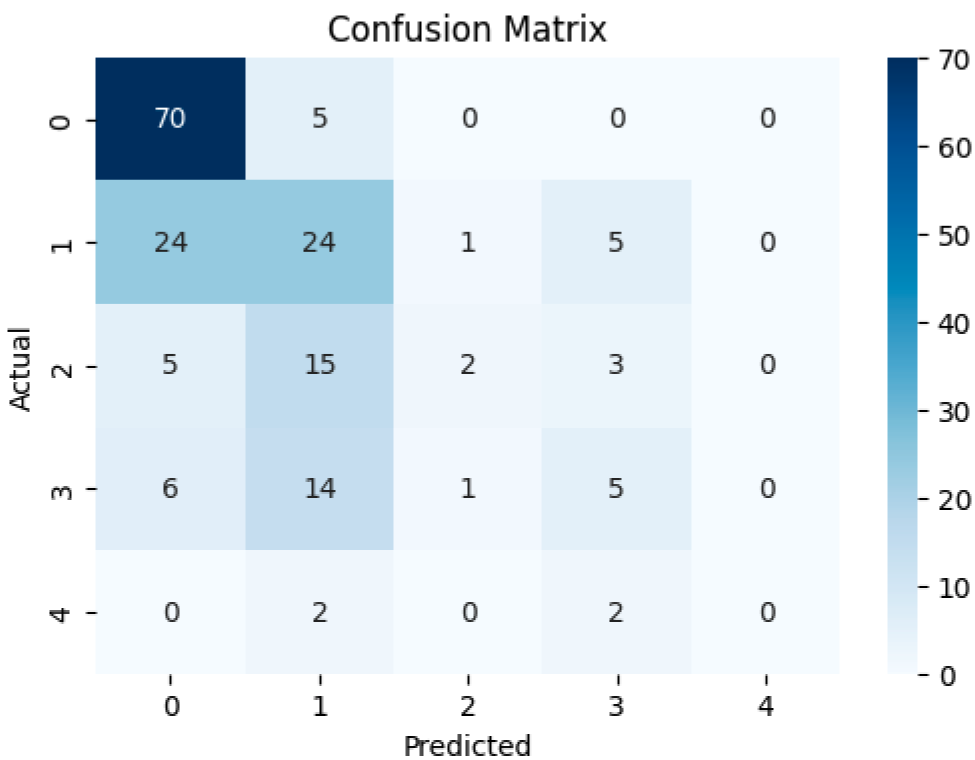


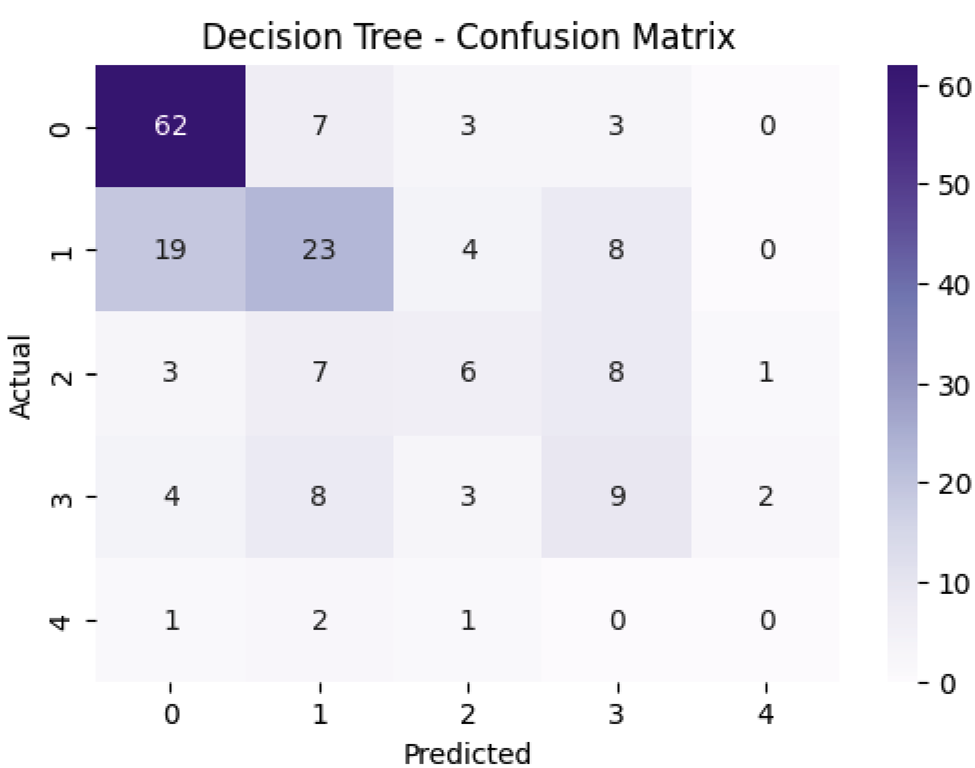
7.2 Classification Results Visualization

- **Confusion Matrix (for Decision Tree and KNN):**

Showed the breakdown of correct vs. incorrect predictions for both classes. It helped us assess whether models were favoring one class over the other.

📌 *Confusion Matrix Screenshots for Both Models*

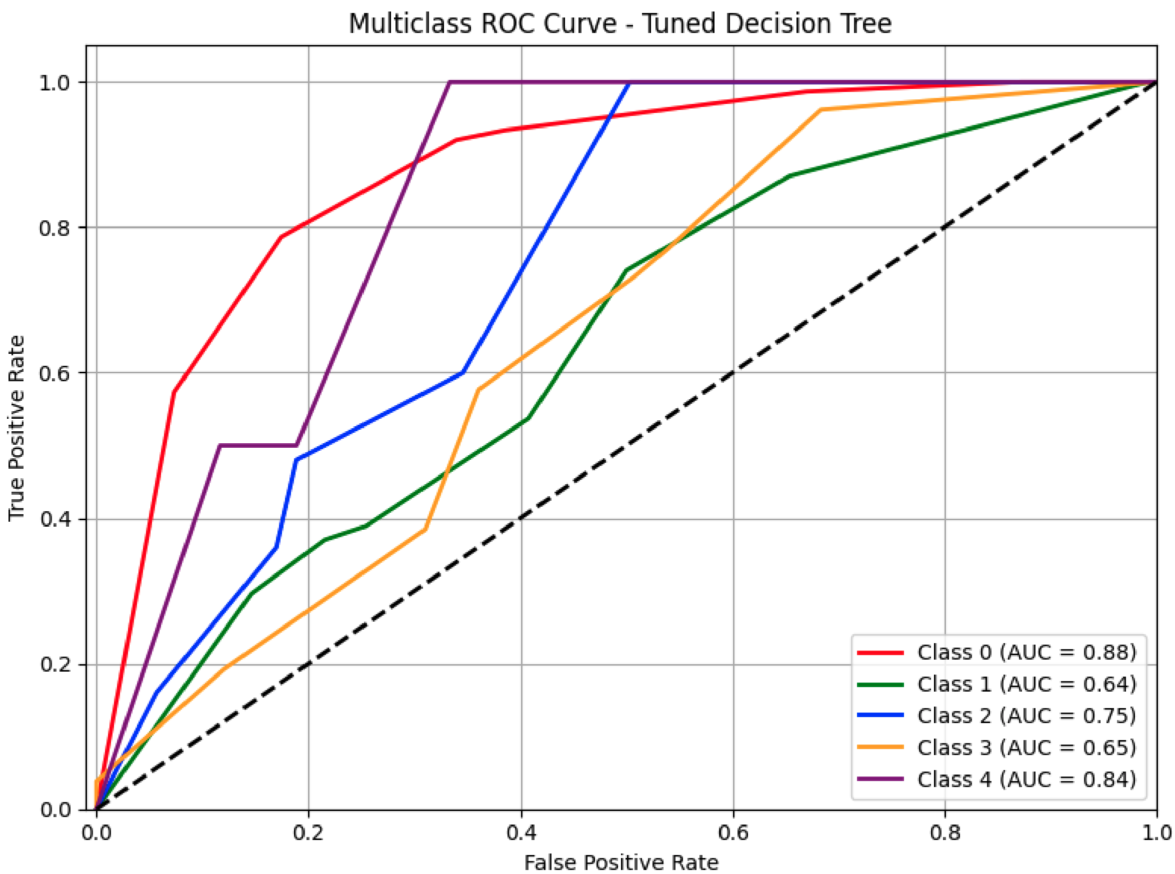

Confusion Matrix
Actual
Predicted
70 5 0 0 0
24 24 1 5 0
5 15 2 3 0
6 14 1 5 0
0 2 0 2 0


Decision Tree - Confusion Matrix
Actual
Predicted
62 7 3 3 0
19 23 4 8 0
3 7 6 8 1
4 8 3 9 2
1 2 1 0 0

- **ROC Curves:**

Demonstrated model performance in terms of sensitivity and specificity. KNN outperformed Decision Tree in ROC-AUC score, indicating a better overall balance between true positive and false positive rates.

📌 *Screenshot of ROC Curve Comparison*



- **Feature Importance Plot (for Decision Tree):**

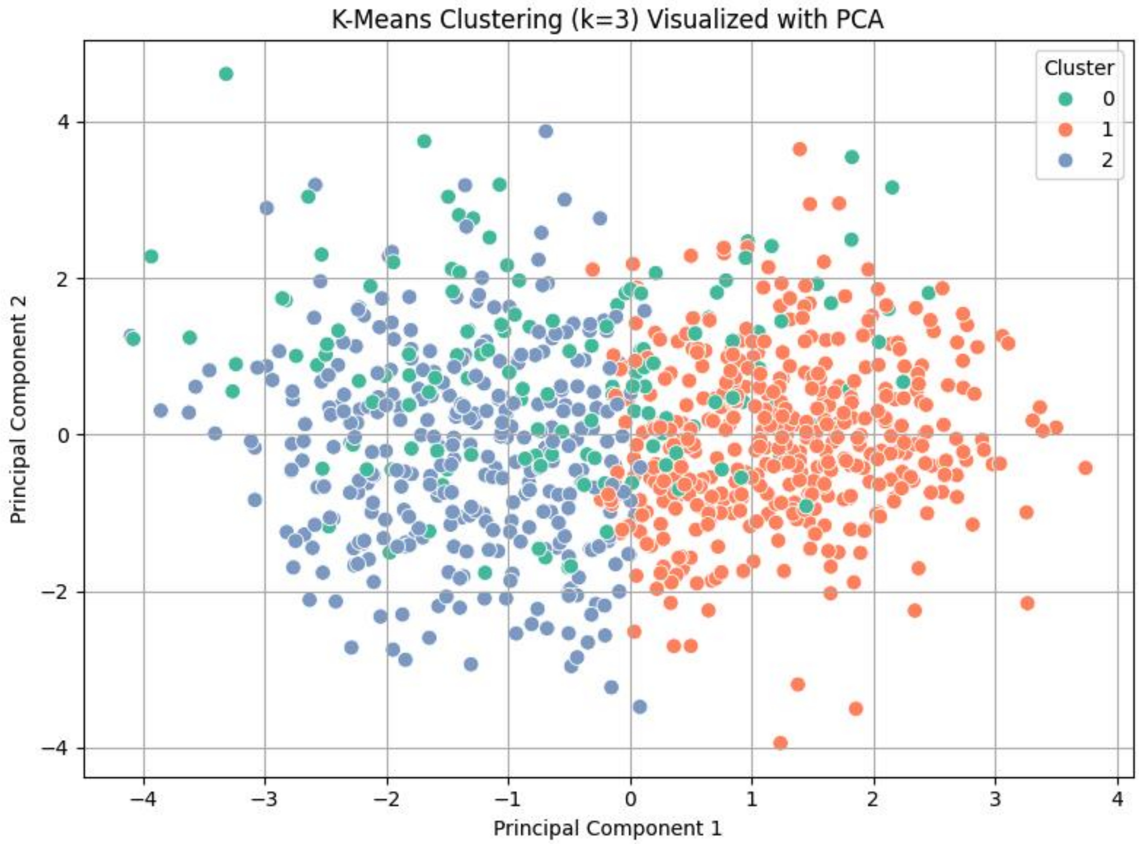
Made it clear which variables were driving the model's decisions. This added transparency and interpretability to our analysis.

7.3 Clustering Visualization

- **K-Means Clusters (Scatter Plot):**

Visualized how the observations were grouped based on thalach, chol, and age. Each cluster suggested a subgroup of patients with similar profiles, potentially helping doctors tailor care.

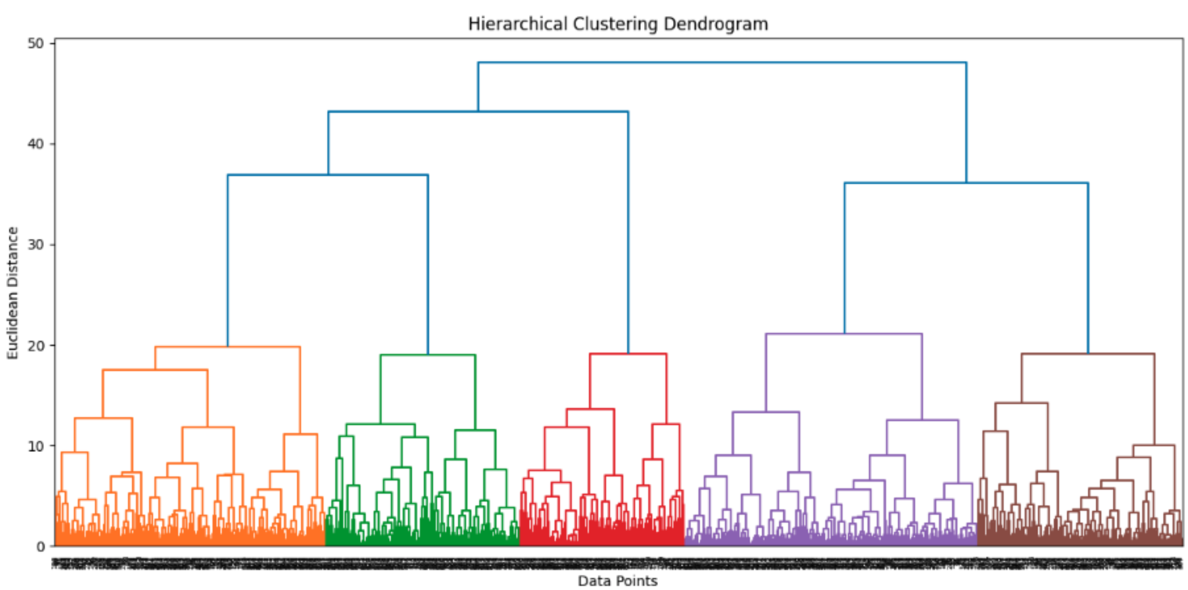
📌 *Screenshot of K-Means Clustering Output*



- **Dendrogram (for Hierarchical Clustering):**

Revealed hierarchical relationships among observations. It helped validate the natural grouping tendency found in the dataset.

📌 *Screenshot of Dendrogram*



7.4 Association Rule Mining

- **Support vs Confidence Scatter Plot:**

Showed how different rules compared in terms of support and confidence. Rules with high confidence and reasonable support were considered most actionable.

8.1 Recommendations for Healthcare Practitioners

- **Risk Stratification:**

By using classification models like KNN and Decision Tree, healthcare providers can build a preliminary screening tool to predict a patient's risk of heart disease based on routine tests (e.g., chest pain type, resting blood pressure, max heart rate).

- **Tailored Intervention:**

The clustering analysis showed that patients can be grouped into distinct subtypes. Each cluster may respond differently to treatment, suggesting a more personalized approach to care planning.

- **Early Warning Rules:**

The association rules uncovered patterns like "If cp = typical angina and thalach > 150, then risk = low" or "If exang = yes and oldpeak > 2.5, then risk = high." These rules can be integrated into clinical decision-support systems for flagging at-risk patients.

8.2 Real-World Applications

- **Mobile Health Monitoring Apps:**

With classification logic embedded, mobile apps could let users self-monitor health metrics and receive real-time risk assessments.

- **Clinical Decision Support (CDS) Systems:**

The mined association rules and predictive models could be embedded in hospital EHR systems to support doctor decision-making at the point of care.

- **Public Health Screening Campaigns:**

The models can assist in designing targeted screening programs by identifying key demographic and clinical attributes that define high-risk groups.

8.3 Next Steps and Scalability

- **Integration with Wearables:**

Features like heart rate and blood pressure from smart devices could feed into our models in real-time, increasing their utility in preventive care.

- **Continuous Model Improvement:**

As more patient data becomes available, retraining and updating models will help maintain accuracy and relevance.

- **Validation on Other Datasets:**

To confirm robustness and avoid dataset-specific bias, models should be validated on external heart disease datasets or real-world hospital data.

9. Ethical Considerations

In any project involving human health data, ethical considerations are not just important—they are essential. Throughout this project, we took deliberate steps to ensure that our work aligned with the core principles of data ethics, including privacy, fairness, transparency, and accountability.

9.1 Data Privacy and Anonymity

- The dataset used (Heart Disease UCI dataset) contains anonymized information, with no personally identifiable details such as names, IDs, or addresses.
- We made sure not to augment the dataset with any external personal data.
- Any sharing or storage of the dataset was limited to academic and research purposes, and all collaborators accessed it securely.

9.2 Fairness and Bias Mitigation

- While the dataset was clean, there were still concerns regarding **representational fairness**. For example, the gender distribution and age groups were slightly imbalanced.
- We analyzed feature distributions and included techniques like stratified sampling in classification tasks to reduce sampling bias.
- Future work should consider using more inclusive datasets from multiple regions and ethnic groups to improve generalizability.

9.3 Model Transparency and Interpretability

- We prioritized the use of interpretable models like Decision Trees and Association Rule Mining so that predictions could be easily explained to stakeholders, especially in clinical contexts where interpretability is crucial.
- Visualizations and rule-based outputs were clearly laid out to ensure transparency in how predictions or recommendations were derived.

9.4 Responsible Use and Limitations

- Our models are meant to **support** healthcare decisions, not **replace** them. No model should be used to make critical medical decisions without human oversight.

- We included disclaimers in our findings to emphasize that this is an academic exercise, and models must undergo regulatory validation and clinical trials before real-world deployment.

9.5 Potential Harm and Mitigation

- Misclassification could lead to either **false assurance** or **unnecessary panic** for patients. For this reason, we highlighted **sensitivity and specificity** in model evaluation and chose conservative thresholds where needed.
- Models are only as good as the data they are trained on. We ensured that preprocessing steps (like outlier removal and feature normalization) were done carefully to avoid introducing artificial patterns.

10. Visualizations Used

Effective visualization was a critical part of our data mining process. It helped us uncover hidden patterns, validate assumptions, and communicate insights clearly. Throughout the deliverables, we used various plots and charts at different stages of the project:

10.1 Exploratory Data Analysis (EDA)

- **Correlation Heatmap:**

Used to identify strong linear relationships between numerical features. For example, features like cp (chest pain type) and thalach (maximum heart rate achieved) showed meaningful correlations with the target variable.

- **Boxplots and Histograms:**

Helped in detecting outliers and understanding the distribution of features like age, chol (cholesterol), and thalach. These visualizations also guided feature scaling and normalization strategies.

10.2 Regression Analysis

- **Scatter Plot (Actual vs. Predicted):**

For the polynomial regression model, a scatter plot was used to compare actual values with model predictions. It provided visual confirmation of model accuracy.

- **Residual Plot:**

Displayed the difference between predicted and actual values. This was useful to confirm that errors were randomly distributed.

10.3 Classification Models

- **Confusion Matrix:**

Presented for each classification model (Decision Tree, KNN) to evaluate true positives, false positives, false negatives, and true negatives.

- **ROC Curve:**

Illustrated the trade-off between sensitivity and specificity. Models with curves closer to the top-left corner indicated better performance.

- **Feature Importance Chart (for Decision Tree):**

Helped in identifying which features had the greatest impact on classification.

10.4 Clustering

- **K-Means Cluster Plot (2D PCA-reduced):**

Showed how the data was segmented into clusters based on similarity.

- **Dendrogram (Hierarchical Clustering):**

Visualized the hierarchical relationships and distances between clusters. Helped determine an appropriate number of clusters.

10.5 Association Rule Mining

- **Support vs Confidence Scatter Plot:**

Displayed interesting patterns found using the Apriori algorithm, showing trade-offs between support and confidence.

- **Sample Rules Table:**

A simple table showing the top association rules, with metrics like support, confidence, and lift.

📌 *Screenshot : Association Rules Table*

Top Association Rules:

	antecedents	consequents	support	confidence	lift
275	(thal_normal, num, restecg_normal, dataset_Hun...	(slope_flat)	0.080435	1.000000	1.406728
369	(dataset_Switzerland, slope_flat, cp_asymptoma...	(num)	0.066304	1.000000	1.807466
210	(num, restecg_normal, dataset_Hungary)	(slope_flat)	0.093478	1.000000	1.406728
493	(restecg_normal, cp_asymptomatic, dataset_Swit...	(num)	0.051087	1.000000	1.807466
388	(num, cp_asymptomatic, dataset_Hungary, thal_n...	(slope_flat)	0.065217	1.000000	1.406728
323	(num, restecg_normal, cp_asymptomatic, dataset...	(slope_flat)	0.075000	1.000000	1.406728
131	(num, dataset_Hungary)	(slope_flat)	0.114130	0.990566	1.393457
146	(restecg_normal, cp_asymptomatic, dataset_Hung...	(slope_flat)	0.105435	0.989796	1.392373
165	(thal_normal, num, dataset_Hungary)	(slope_flat)	0.100000	0.989247	1.391602
196	(thal_normal, restecg_normal, cp_asymptomatic,...	(slope_flat)	0.094565	0.988636	1.390742